

Benchmarking of Switching Technologies under Various Traffic Distributions in Pan-European and US WDM Core Networks

Thierry Zami⁽¹⁾, Ihtesham Khan⁽²⁾, Jesse Simsarian⁽²⁾

⁽¹⁾ ASN/Nokia France, thierry.zami@asn.com

⁽²⁾ Nokia Bell Labs, Murray Hill, NJ, USA

Abstract We review the conditions of network topology and traffic distribution under which pure fiber switching can accommodate a total capacity close to wavelength switching for a North American and a Pan-European WDM networks ©2026 The Author(s)

Introduction

As the mean spectral efficiency in WDM core networks is now close to Shannon's limit, deploying numerous parallel fibers per link is an approach presently being implemented to keep increasing substantially the total maximum network capacity (MNC) in coming years [1-5]. This should be done while still lowering cost per transmitted Gbit/s. To that aim, pure fiber switching (FiS) without wavelength switching (WaS) complexity [6-7] is envisaged, possibly in complement of pluggable coherent interfaces [8] and "multirail" optical amplifiers [9]. However, FiS effectiveness strongly depends on the traffic exchanged between the network nodes. In that context, this paper compares FiS vs. WaS and vs. opaque switching (OpS), i.e., by routing all the signals in the electrical layer at every network node [10]. This benchmarking is performed for 2 core network topologies and various traffic conditions to understand whether and how FiS could be at least as efficient as WaS in terms of total served throughput.

Network planning assumptions

WaS, FiS and OpS should be simulated with their specific strengths and limitations for a fair comparison. Hence, current electrical cross-connects for OpS cannot handle more than a few hundred Tb/s to optimally fill each optical channel capacity via traffic regrooming at each node. The most widespread optical cross-connect (OXC) setup for dynamic WaS in backbone networks resorts to wavelength selective switches (WSS) with up to 40 external and internal fiber ports at most [11]. Hence, the "route&select" WSS-based OXC [12] could be stacked to handle situations beyond 4 links (or directions) per OXC and 5 WDM fibers/link. On the other hand, FiS can rely on fiber

cross-connects (FXC) with hundreds of input/output fibers [13]. Consequently, one FXC can serve up to several thousand Tb/s transit traffic with unmatched compactness and electrical power efficiency per Gb/s. However, FiS routing granularity in Gb/s is substantially higher than the OpS and WaS ones. So, it could be harder to fill efficiently if the traffic breakdown is inappropriate.

Our WDM network planning accounts for these issues with N bidirectional pairs of fibers/link and N up to 40. We consider EBN39 and NFSNET core WDM networks illustrated in Fig. 1, respectively made of 14 and 39 nodes, as well as 21 and 59 WDM links modelled with standard single mode fibers having 0.2 dB/km loss and 80 km longest spans. Fiber spans are interleaved with Erbium doped fiber amplifiers compensating for their loss with 5.5 dB noise figure over the 4800 GHz-wide C-band. We simulate transmission of 32 channels 150 GHz apart per fiber, emitted and received by 138 GBaud embedded elastic optical transponders (EOT). Channels are either routed by electrical cross-connect ("OpS" case) or by N stacked WSS-based OXCs at each network node ("WaS" case) or directly by FXC ("FiS" case). Only one of these 3 switching technologies is applied at a time. Each channel can transport from 400 to 1200 Gb/s with steps of 200 Gb/s by means of probabilistic constellation shaping, depending on its final electrical signal to noise ratio (SNR). We calculate this SNR by considering the physical degradations added to the signal when crossing network elements along its light path (LP), like imperfections of pair of emitting and receiving EOTs; amplified spontaneous emission from optical amplifiers; non-linear effects from WDM transmission in fibers; optical filtering, crosstalk and polarization dependent loss from optical routing devices.

The simulated traffic called $T(x)$ means that bidirectional symmetrical connections between pairs of nodes are randomly requested for x , $2 \times x$, $3 \times x$ or $4 \times x$ Gb/s throughput per direction with uniform 0.25 probability. The first part of this paper assumes uniform geographical distribution of $T(x)$,

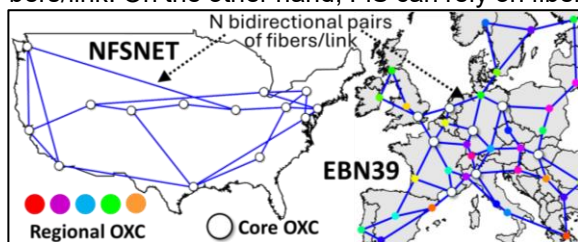


Fig. 1: NFSNET [14] and EBN39 WDM network topologies

randomly over all the pairs of network nodes. Optical transparent connections under WaS and FiS are transmitted without intermediate traffic regrooming nor optoelectronic regeneration, with end-to-end electrical grooming of traffic and inverse multiplexing possibly over several optical carriers following different LPs. OpS reaggregates traffic in the intermediate electrical nodes crossed along the channel LP. We simulate the incremental allocation of $T(x)$ services along the shortest LP available between the requested source and destination nodes for each demand. If our routing algorithm can find a LP with enough residual spectral resources, they are allocated to this demand. Otherwise, the demand is rejected. Each network simulation runs until 1% of the total demanded capacity has been discarded. At this point, we establish MNC and the required number of EOTs per Gb/s (N_{EOT}). For fair benchmarking of different configurations based on distinct numbers of fibers/link N , MNC and N_{EOT} values reported in this paper are the ones divided by N and averaged with 100 different draws of $T(x)$ traffic, each one with a distinct random seed.

Simulation results for NFSNET

Fig 2.a, b and c show the evolution of MNC/fiber obtained for NFSNET for 8, 24 and 40 fibers/link and for $T(x)$ with x from 1600 to 12800 Gb/s. Some of these traffics have connections carrying more than 20 Tb/s which can exceed the capacity of one fiber along some LPs. Several trends appear in fig. 2. Unsurprisingly, OpS leads to much higher MNC than WaS or FiS for any simulated traffic. However, this is at the cost of 53 % larger

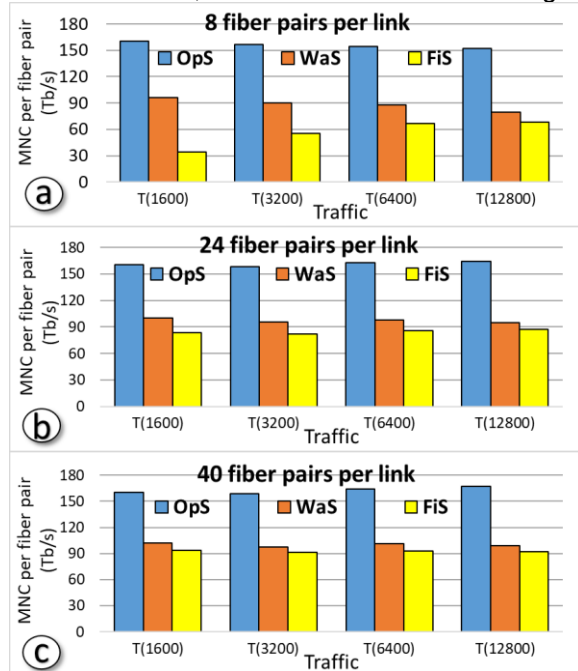


Fig. 2: Comparison MNC per fiber pair for OpS, WaS and FiS applied to NFSNET with different traffics $T(x)$

N_{EOT} than WaS when $N=40$ (not reported in fig. 2). Secondly, while MNC/fiber is stable under OpS for any traffic, it drops for WaS when the mean exchanged traffic grows, especially if N is low. This results from more spectral fragmentation under WaS if connections need more parallel carriers. This fragmentation is more difficult to circumvent without optoelectronic regeneration with relatively low N . By contrast, this effect disappears with $T(12800)$ and $N=40$ in fig. 2c. Unlike WaS, MNC/fiber substantially grows under FiS along with the mean exchanged capacity per connection if N is low, i.e., $N=8$. Because 1% blocking can arise quite early along traffic growth without enough fibers/link to interconnect all the required pairs of nodes, while fiber connections already established are not yet sufficiently filled. This notably affects MNC when the throughputs of the individual demands are far from filling the capacity of one single fiber, i.e., with $T(1600)$ and $N=8$, even throughout the longest NFSNET LPs. Once one fiber connection has been established between one pair of nodes, end-to-end grooming can then more efficiently and gradually fill it. Hence, this downside of low exchanged traffic does not occur with FiS if N is high enough to serve at least once each pair of nodes exchanging traffic, as shown in fig. 2c where $N=40$ and then MNC/fiber remains stable over the range of simulated $T(x)$ for any switching technology.

Fig. 2 also shows that FiS in NFSNET does not yield greater MNC than WaS. So, by optimally implementing each optical switching technology, the "FiS" MNC can approach the "WaS" one (see fig. 2 with $T(12800)$), but cannot exceed it. This is because even when the whole capacity of individual connections occupies a wide fraction of the C-band, inverse multiplexing makes WaS more flexible. On the other hand, FiS is expected to have a lower cost per Gb/s than WaS which requires stacking many WSS-based OXCs in WDM networks with $N>10$ [6]. But this argument better holds only if their respective MNCs remain close. Fig. 2 illustrates that the most important criteria to reach that goal is the high enough number of fibers/link. In practice, [7] showed higher MNC with FiS than with WaS if connections are demanded as an integer number of full fiber capacity.

Simulation results for EBN39

In this section, we examine whether the FiS optimal conditions for EBN39 are like the NFSNET ones by focusing on a smaller set of configurations reported in fig. 3a and 3b. The main trends remain the same but with the following noticeable difference: unlike NFSNET, there is a more important reduction of MNC/fiber between WaS and FiS for $T(3200)$, even with the highest number of

fibers/link $N=40$. So, if the traffic is not large enough, $N=40$ is not sufficient to make FiS equivalent to WaS in EBN39. This stems from the much larger number of equiprobable couples of network nodes to be interconnected in EBN39 than in NFSNET. This illustrates how FiS effectiveness strongly depends on network topology and on the geographical distribution of traffic. This dependency appears differently if simulating EBN39 with a hierarchical traffic $T(H)$ defined as follows: 33.3% of connections carry 6.4 or 12.8 Tb/s and are requested between random pairs of core nodes defined in fig. 1, while 66.6 % of them carry 3.2 or 6.4 Tb/s and are demanded between a randomly picked regional node and any of its 3 closest core nodes. Fig. 4 depicts the results under that condition for $N=8$, 24 and 40, to be directly compared against the performance in fig. 3 with uniform geographical traffic $T(3200)$. $T(H)$ partly closes the MNC gap between OpS and WaS since the drop of MNC/fiber with $N=40$ is only -36% with $T(H)$ vs. down to -57% with uniform traffic (c.f. fig. 3b). Another noticeable difference with $T(H)$ is that from $N=24$, MNC/fiber no longer meaningfully differs from WaS to FiS, unlike under uniform traffic $T(3200)$. Therefore, despite notably more nodes in EBN39 than in NFSNET, MNC performances of WaS and FiS in EBN39 with $T(H)$ become similar to the NFSNET ones. Because what matters for FiS efficiency is not the intrinsic number of nodes in the simulated topology, but its maximum quantity of nodes possibly interconnected. In EBN39, this number

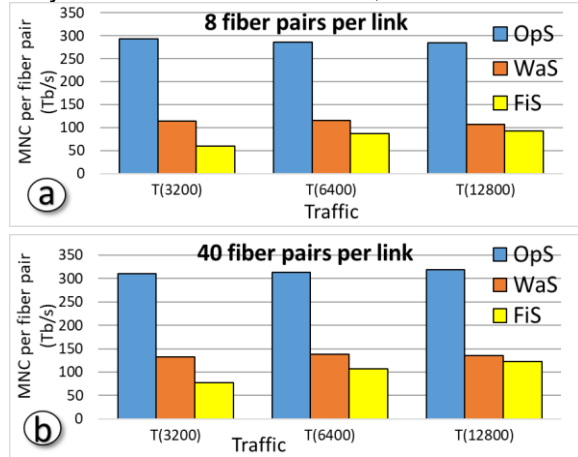


Fig. 3: Compared MNC per fiber pairs for OpS, WaS and FiS applied in EBN39 with different traffics $T(x)$

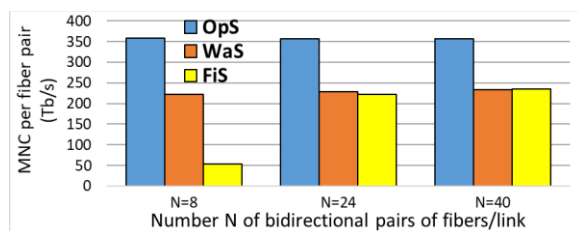


Fig. 4: MNC per fiber pairs for OpS, WaS and FiS applied in EBN39 for the same hierarchical traffic distribution $T(H)$

Table 1: N_{EOT} for EBN39 with 40 fibers/link; comparing 2 traffics with similar mean exchanged capacity per connection

	$T(3200)$	$T(H)$
OpS	6.7 per 800Gb/s	4.25 per 800Gb/s
WaS	2.6 per 800Gb/s	2 per 800Gb/s
FiS	2.6 per 800Gb/s	2 per 800Gb/s

decreases from up to 741 under uniform traffic down to no more than 121 under the $T(H)$ traffic we have described.

Table 1 reports N_{EOT} for EBN39, showing the same lowest optimal values for WaS and FiS when $N=40$. With $T(3200)$ (resp. $T(H)$), OpS leads to 159% (resp. 114%) extra N_{EOT} as compared to the 2 optical switching setups. This percentage difference is due to the longer LPs with $T(3200)$ uniform traffic crossing more intermediate nodes and then needing more EOTs for OpS.

Discussion and conclusion

We have implemented a WDM network planning tool with persistent wavelength routing effectiveness up to 40 fibers/link thanks to comprehensive inverse multiplexing. It exemplifies the conditions under which FiS can compete against WaS in terms of mean capacity/fiber for 2 core WDM topologies. Hence, to be competitive with very flexible high-performance transponders, FiS should be applied to networks having a large number of fibers/link and nodes exchanging high throughput. But these obvious conditions might not be fulfilled in most networks in a short time range. Hence, even under non-ideal traffic or infrastructure, it is interesting to see how MNC with FiS can still be close enough to MNC with WaS to maximize the cost benefit per Gb/s from FiS.

In that respect, we have shown that whereas a uniform traffic distribution might be detrimental for FiS, especially for the network topology exhibiting many nodes, hierarchical traffic distribution could still foster FiS for these large topologies. However, as long as traffic breakdown remains far from $T(6400)$ -like mean exchanged capacity and with number of fibers/link lower than 10, pure FiS will not be effective enough (see fig. 2, 3 and 4 for $N=8$) in core WDM networks having more than 20 nodes. The most valuable all-optical approach would then likely consist of combined WaS and FiS-capable OXCs [15]. It is also noteworthy that for both tested networks, OpS can transport from 30% to 60% more network capacity per fiber than full transparent WaS or FiS at the cost of many more transponders needed per Gb/s.

Nevertheless, despite the "one fits all" switching layout not yet being clearly established for WDM networks, the recent announcement of the future "full band transponders" [16] further paves the way towards effective FiS since these transponders will fill a whole fiber all at once.

References

- [1] S. L. Woodward, "ROADM options in optical networks: flexible grid or not?", Proc. OFC 2013, paper OTh3B.1, OFC/NFOEC'2013, March 2013, <https://doi.org/10.1364/OFC.2013.OTh3B.1>
- [2] R. Schmogrow, "Solving for Scalability From Multi-Band to Multi-Rail Core Networks", in Journal of Lightwave Technology, vol. 40, no. 11, pp. 3406-3414, June 2022, doi: 10.1109/JLT.2022.3172628
- [3] K. Sato, Y. Mori, H. Hasegawa, and K.-I. Sato, "Algorithm for Raising OXC Port Count to Meet Traffic Growth at Minimum Cost", Journal of Optical Communications and Networking, Vol 9, n°2, A263-A270, February 2017, <https://doi.org/10.1364/JOCN.9.00A263>
- [4] Masahiko Jinno, "Spatial Channel Network (SCN): Opportunities and Challenges of Introducing Spatial Bypass Toward the Massive SDM Era", Vol 11, n°3, pp. 1-14, March 2029, <https://doi.org/10.1364/JOCN.11.000001>
- [5] T. Zami, I. Khan, J. Simsarian, M. Szczerban, D. Neilson and R. Ryf, "How "Pay as You Grow" OXC Stacking Affects the Performance of Wavelength-Routing SDM/WDM Transparent Networks", Paper M.03.06.4 ECOC'2025, October 2025, doi:10.1109/ECOC66593.2025.11263208
- [6] M. Szczerban, T. Zami, S. Bidkar, C. Kelly, D. Neilson, R. Ryf, and J. Simsarian, "Fibre cross connect vs. stacked-WDM capacity and cost assessment in fibre-rich optical core networks", Paper W1E.5, ECOC'2024, September 2024
- [7] I. Khan, M. Szczerban, T. Zami, S. Bidkar, C. Kelly, D. Neilson, R. Ryf, J. Simsarian, "Networking Analysis of Fiber Cross-Connect and Stacked-WDM Architectures in Bandwidth Abundant Optical Networks", Paper P2, ONDM'2025, May 2025
- [8] E. Pincemin, A. Gueutier, O. Renais, F. Herviou, Y. Delisle, M. Bluethner, I. Riggs, A. Malik, T. Nielsen, T. Williams, J. Larikova, O. Bertran Pardo, R. Kapuscinski, A. Kakkar, M. Zontini, G. Mussi, A. Gugliera, B. Sleiman, "Multi-Vendor Interoperable 800G Optical Communications Over 1602-km Through 800G-OpenROADM, 800-ZR and 800GBASE-2xFR4 Fiber Links", Paper Th4B.6, OFC'2026, March 2026
- [9] COHERENT, "Enabling efficient capacity scaling through multi-rail photonic technology", Presentation in ECOC'2025 Market Focus session, October 2025
- [10] J. Rahn, E. Torrenco and M. Mitchell, "IP Over DWDM at Scale: Pluggable Transformation at Meta", Paper Th.02.06.4, ECOC'2025, October 2025, doi: 10.1109/ECOC66593.2025.11263368
- [11] M. Iwama, M. Takahashi, M. Kimura, Y. Uchida, J. Hasegawa, R. Kawahara, and N. Kagi, "LCOS-based Flexible Grid 1x40 Wavelength Selective Switch Using Planar Lightwave Circuit as Spot Size Converter", Paper Tu3A.8, OFC 2015, March 2025, <https://doi.org/10.1364/OFC.2015.Tu3A.8>
- [12] B. Collings, "New devices enabling software-defined optical networks", in IEEE Communications Magazine, vol. 51, n°3, page 66-71, March 2013, DOI: 10.1109/MCOM.2013.6476867
- [13] R. Ryf, J. Kim, J. P. Hickey, A. Gnauck, D. Carr, F. Pardo, C. Bolle, R. Frahm, N. Basavanthally, C. Yoh, D. Ramsey, R. Boie, R. George, J. Kraus, C. Lichtenwalner, R. Papazian, J. Gates, H. R. Shea, A. Gasparyan, V. Muratov, J. E. Griffith, J. A. Prybyla, S. Goyal, C. D. White, M. T. Lin, R. Ruel, C. Nijander, S. Arney, D. T. Neilson, D. J. Bishop, P. Kolodner, S. Pau, C. Nuzman, A. Weis, B. Kumar, D. Lieuwen, V. Aksyuk, D. S. Greywall, T. C. Lee, H. T. Soh, W. M. Mansfield, S. Jin, W. Y. Lai, H. A. Huggins, D. L. Barr, R. A. Cirelli, G. R. Bogart, K. Teffeau, R. Vella, H. Mavoori, A. Ramirez, N. A. Ciampa, F. P. Klemens, M. D. Morris, T. Boone, J. Q. Liu, J. M. Rosamilia, and C. R. Giles, "1296-port MEMS Transparent Optical Crossconnect with 2.07Petabit/s Switch Capacity", paper PD28, OFC'2001, March 2001, <https://doi.org/10.1364/OFC.2001.PD28>
- [14] E. He, X. Wang, V. Vishwanath and J. Leigh, "AR-PIN/PDC: Flexible Advance Reservation of Intradomain and Interdomain Lightpaths", Globecom 2006, November 2006
- [15] C. Blaizot, E. Dotaro, L. Noirie and A. Jourdan, "Multi-granularity optical networks", pp 1-20, ONDM'2000, February 2000
- [16] S. Gareau, D. Perras, F. Chang and T. Norman, "The Role of Pluggables vs. Embedded Transponders in Optical Networks", Paper W3A1, OFC'2026, March 2026